2313666 殷家兴 自动化 现控⑥

1. 解: $n=3$ $A=\begin{pmatrix}-1&0&0\\0&-2&-3\\1&0&1\end{pmatrix}$ $B=\begin{pmatrix}1&0\\0&1\\0&-1\end{pmatrix}$ $C=\begin{pmatrix}1&0&0\\0&1&1\end{pmatrix}$

~~判断能控性:~~

$$\cancel{U=[B \vdots AB \vdots A^2B]=\begin{bmatrix}1&0&-1&0&1&0\\0&1&0&1&-3&1\\0&-1&1&-1&0&-1\end{bmatrix}}$$

$c_1B=(1\ 0)$ $r_1=1$ $c_2B=(0\ 0)$

$c_2AB=(1\ 0)$ $r_2=2$

故 $E=\begin{bmatrix}c_1B\\c_2AB\end{bmatrix}=\begin{pmatrix}1&0\\1&0\end{pmatrix}=\begin{pmatrix}1&0\\0&0\end{pmatrix}$ ~~rank~~ $\mathrm{rank}(E)=1<2$

因此系统无法用状态反馈解耦

2. 解: 由状态方程解出 $u=\dot{x}+x$, 代入 $J(u)$ 得到:

$$J=\frac{1}{2}\int_0^1 (3x^2+(\dot{x}+x)^2)\,dt$$

$$F=\cancel{\frac{1}{2}3}\ \frac{3}{2}x^2+\frac{1}{2}(\dot{x}+x)^2$$

已知边界条件 $x(0)=1$ $x(1)=0$, 则极值轨线满足:

$$\frac{\partial F}{\partial x}-\frac{d}{dt}\left(\frac{\partial F}{\partial \dot{x}}\right)=3x+(x+\dot{x})-\frac{d}{dt}(x+\dot{x})=0$$

即 $\cancel{4x-\ddot{x}=0}$ $\ddot{x}-4x=0$

则 x 使 $J(x)$ 达到极小值的必要条件为:

$$\begin{cases}\cancel{4x-\ddot{x}=0}\quad \ddot{x}-4x=0\\ x(0)=1\\ x(1)=0\end{cases}$$

$x(t)=c_1e^{2t}+c_2e^{-2t}$

又 $x(0)=1$ $x(1)=0$

得 $\begin{cases}c_1=\dfrac{1}{1-e^4}\\ c_2=\dfrac{-e^4}{1-e^4}\end{cases}$ 即 $x(t)=\dfrac{e^{2t}}{1-e^4}+\dfrac{-e^4\cdot e^{-2t}}{1-e^4}$

$$u(t)=x(t)+\dot{x}(t)=\frac{e^{2t}}{1-e^4}+\frac{-e^4\cdot e^{-2t}}{1-e^4}+\frac{2e^{2t}}{1-e^4}+\frac{2e^4\cdot e^{-2t}}{1-e^4}$$

$$=\frac{3}{1-e^4}e^{2t}+\frac{e^4}{1-e^4}e^{-2t}$$

3. 解: 由 $\dot{x}=x+u$ 可得 $u=\dot{x}-x$，代入 $J(u)=\frac{1}{2}\int_0^1(x^2+u^2)dt$ 得

$$J=\frac{1}{2}\int_0^1\left(x^2+(\dot{x}-x)^2\right)dt$$

$$f=\frac{1}{2}x^2+\frac{1}{2}(\dot{x}-x)^2$$

则极值轨线满足欧拉方程

$$\frac{\partial f}{\partial x}-\frac{d}{dt}\left(\frac{\partial f}{\partial \dot{x}}\right)=x+(x-\dot{x})-\frac{d}{dt}(\dot{x}-x)=0$$

即 $\ddot{x}-2x=0$

于是使 x 使 $J(x)$ 达极小值的必要条件为

$$\begin{cases}\ddot{x}-2x=0\\ x(0)=1\\ [\dot{x}-x]_1=0\end{cases}$$

通解为: $x(t)=c_1e^{\sqrt{2}t}+c_2e^{-\sqrt{2}t}$

又 $x(0)=1$，$[\dot{x}-x]_1=0$

得
$$\begin{cases}c_1=\dfrac{\sqrt{2}+1}{\sqrt{2}+1+(\sqrt{2}-1)e^{2\sqrt{2}}}\\ c_2=\dfrac{\sqrt{2}-1}{\sqrt{2}-1+(\sqrt{2}+1)e^{-2\sqrt{2}}}\end{cases}$$

即 $x(t)=\dfrac{\sqrt{2}+1}{\sqrt{2}+1+(\sqrt{2}-1)e^{2\sqrt{2}}}e^{\sqrt{2}t}+\dfrac{\sqrt{2}-1}{\sqrt{2}-1+(\sqrt{2}+1)e^{-2\sqrt{2}}}e^{-\sqrt{2}t}$

$$\dot{x}(t)=\frac{2+\sqrt{2}}{\sqrt{2}+1+(\sqrt{2}-1)e^{2\sqrt{2}}}e^{\sqrt{2}t}+\frac{\sqrt{2}-2}{\sqrt{2}+1+(\sqrt{2}+1)e^{-2\sqrt{2}}}e^{-\sqrt{2}t}$$

$$u(t)=\dot{x}(t)-x(t)=\frac{1}{\sqrt{2}+1+(\sqrt{2}-1)e^{2\sqrt{2}}}e^{\sqrt{2}t}-\frac{1}{\sqrt{2}-1+(\sqrt{2}+1)e^{-2\sqrt{2}}}e^{-\sqrt{2}t}$$